

Team Gary Design Outcomes Report

Introduction

Need Statement

UBC students experiencing food insecurity face challenges related to food access, financial pressure, or both. Our design focuses on two main groups: students with busy schedules who struggle to find food at the time and location they need, and students who rely on food assistance programs and require timely support without added stress, stigma, or complexity.

Design Requirements

To address these needs, we defined five key design requirements:

1. **Intuitive navigation**

The interface must be easy to learn and use, allowing students to quickly find core features without confusion.

2. **Efficient filtering interaction**

The system must let users smoothly adjust preferences (e.g., price, distance, dietary needs) and immediately see relevant food options.

3. **Real-time food updates**

The system must display up-to-date information about item availability, special deals, and when food is sold out.

4. **Privacy control**

Users must be able to control what personal information (e.g., name, budget, address) is visible to others to reduce stigma and protect confidentiality.

5. **High visibility of information**

Information about food options, locations, and pickup details must be presented in a clear, readable, and easily scannable format.

Initial Design Approach

We chose an alternative food-assistance approach that replaces traditional, in-person food bank visits with self-service pickup via vending machines. Instead of waiting in line and selecting items in a public setting, UBC students can register, complete eligibility surveys, and choose food on their mobile phones, then pick up their selected items from a campus vending machine at a convenient time. This approach is intended to reduce time pressure for busy students and lower the stigma and stress often associated with food banks.

Refined Design Approach

Initially, our approach focused on a kiosk-centric experience, where students would use a large display on the vending machine to complete surveys and choose items onsite. Based on feedback and usability concerns (time pressure, stigma, and crowding around the machine), we refined the approach to a mobile-based model, where the vending machine is used primarily for pickup and most interaction happens in the app. In addition, we decided to integrate both regular low-cost grocery options and food-bank items into the same interface so that using the system does not visibly mark users as “food bank users,” further reducing stigma.

Conceptual Model Iterations



Initial Conceptual Model

Our initial conceptual model treated the system as a hybrid between a food-bank service and a smart vending machine. Students created an account and completed basic registration in the app, but the vending machine remained the central point of interaction: users approached the machine, authenticated themselves, and then browsed, selected, and confirmed items directly on the large screen before immediately receiving their food. Conceptually, the machine acted as both the main interface and the point of fulfillment, with the app playing a supporting role for account setup.

Refined Conceptual Model



As we refined the design, our conceptual model shifted toward an Amazon-locker-style pickup service that combines **grocery ordering with food assistance**. In the revised model, the mobile app is the primary interface where students register, adjust their preferences, browse both regular and subsidized items, build a cart, and schedule a pickup time and location. Each order is linked to a unique QR code or PIN that represents the student's reservation, and the vending machine is now conceptualized purely as a pickup endpoint: students arrive at their chosen time, authenticate using their code, and collect their pre-reserved items without needing to make choices in public or spend additional time at the machine.

Developed through cognitive walkthrough

Findings from our cognitive walkthrough led to several important refinements within this conceptual model. First, we clarified navigation and labels after realizing that users did not always understand the difference between "Ordering" and "Food Bank." We revised menu labels and added a brief explanation on the home screen so that students can clearly see that they can either order regular groceries or access food-bank resources within the same app. Second, we strengthened feedback for key steps: actions such as adding items to the cart and confirming pickup time now trigger clear visual feedback and are supported by prominent

“Confirm order” button, followed by an “Order Review” page where users can verify items, location, and time before finalizing. Finally, we improved pickup visibility by clarifying the UBC map and locker status, and by adding reminders to the confirmation page, reinforcing the mental model that an order is reserved for a specific time slot at a specific locker.

Evaluation and Prototyping Methods (2 pages max)

Research questions

1) To what extent can first-time student users complete the full flow (browse → add items → choose locker/time → confirm) without assistance?

2) Do the filtering and selection controls let users efficiently narrow to the foods, pickup time, and locker they want with minimal interaction cost, and which control types (category filters, search, time slot list, map) create the most friction?

Question 1) is important to answer as a part of our usability testing. Our target users (students facing financial food insecurity or time constrained insecurity) are often under stress from their lack of food access and demanding schedules. So, we made it a goal throughout the design process that the interface should have good intuitive navigation or intuitive flow. If the interface was not easy to use for a first-time user, then it would have failed to provide support without added stress mentioned in our user needs. **Question 2)** Since our users often have specific constraints (food type, or time constraints), the interface must allow them to locate items that fit their needs. If the controls for filtering and selection are not well implemented, users are likely to abandon the task and not use our interface to combat their food insecurity.

Summary of evaluation

The final usability test used a Think-Aloud observation paired **with** a semi-structured interview to evaluate the prototype's flow and satisfaction. The protocol first included a 5-minute general observation where participants freely used the app as they liked. Then, we assigned participants a task, and they had to complete the task. We documented any points of interest on a coding sheet. Then, we asked them questions about using the interface and documented their answers. After the study sign-off, we changed the research questions to be specific to the limited implementations that our interface has.

Evaluation rationale

Justification of the evaluation questions: The first research question examines the holistic user experience, including emotional response and navigational competence. Specifically, this involves observing the user's ability to operate the interface with minimal supervision, the intuitiveness of task execution, and their emotional reactions throughout the process. Then, the second research question targets specific design elements that require optimization.

Our two research questions serve as complementary functions together. The first evaluates the overall strengths and weaknesses of the *current* interface regarding usability and sentiment, while the second one pinpoint actionable area for improvement in the *future* design phase.

Justification of evaluation methods and protocol: The first research question is addressed via think-aloud observation. This research question focuses on the user's feelings, and thought process. By observing the participants' actual navigation process and listening to their verbal feedback, we collect both *objective* behavior and *subjective* emotional responses to effectively answer this research question.

The second research question identifies areas of improvement in our UI, primarily through the post-task interview. We observe friction points during tasks and discuss them in semi-structured interviews to uncover the underlying reasoning. This flexible approach allows us to capture diverse opinions and understand exactly why specific features fail.

Our protocol allows us to triangulate our data by validating behavioral actions against verbal reasoning. Utilizing critical incident logging methods to capture key user sentiments ensures our analysis accurately reflects the participants' genuine struggles and mental models without researcher bias.

Prototyping approach and rationale

Prototype Objectives: Our primary prototyping objective was to ensure that 80% of test users could complete the ordering workflow in under five minutes without assistance, as well as, making all users feel inclusive. We value efficiency, utilizing consistent styling and highly visible buttons to test if a clean aesthetic could successfully minimize distractions and facilitate intuitive wayfinding. We also aiming to design an interface that allows users to access food resources without fear of social stigma. To achieve this, we adopted a horizontal prototyping approach that covered the full user journey for both the "Regular Grocery" and "Food Bank" streams. This scope was necessary to demonstrate the application's dual-ecosystem model, showing how

revenue from regular orders supports the food bank. Both streams lead to a QR code acting as a token to pick up food orders at a locker, thereby achieving inclusivity.

Prototyping Process and Methods: We began with low-fidelity sketches, initially planning to realize only the food bank stream in the Figma prototype, but eventually expanded to cover both user streams for conceptual clarity. After internal piloting, several refinements were made. We renamed the confusing “Food Bank Pick Up” label to “Food Bank Ordering” on the Welcome Page and added an Apple Pay screen to complete the transaction logic.

Lesson learned: Overall we received pretty positive feedback on our prototypes such as clean design and smooth flow. However, we also discovered that omitting small details, such as having repeated items, no quantity selectors in the add-to-cart logic, or pages not automatically advancing after payment, could potentially disrupt the user experience. We initially viewed these as extra polish, but they proved critical for usability, teaching us that an effective medium-fidelity prototype must support the complete interaction loop, not just the visual layout.

Analysis (1 pages max)

Research Question 1: To what extent can first-time student users complete the full flow (**browse** → **add items** → **choose locker/time** → **confirm**) without assistance?

Claim: Almost all participants in our study could complete the full flow and reported a positive experience roughly 8-9/10 ([plot 1](#)), but many found minor confusion at specific steps.

Evidence:

1. In the post-task interview, 8 of 10 participants said "yes" because they knew what to do when they opened the app for the first time.
2. 6 users rate the experience between 8 and 10 on a 10-point scale with a mean of about 8.5 ([plot 1](#)). 4 participants used positive terms such as "pretty good" and "pretty straightforward," which are consistent with high satisfaction.
3. 70% of the participants think that it **wasn't clearly visible when items were added to the checkout cart**. ([plot 2](#)) And causing the process to be unsmooth.
4. 2 participants were unsure whether they were **ordering as food bank users or donating**, which made them stop and think.

We are moderately confident about this claim” Users are satisfied with the app with minor confusion” because

- Strength:
 - The quantitative ratings and yes/no questions are consistent with the qualitative comments
 - We asked their thoughts of whether to be fast or not, rather than using the time to capture the individual difference.
- Limitation:
 - We introduced the project and **explained the purpose of the app at the beginning of the study**. This "first glance" may have helped the participants complete the flow **more easily than** new users exploring the app alone.
 - The ratings may be **biased upward** due to social desirability: because the sessions were **moderated** in person, some participants may have **felt uncomfortable giving very low scores or criticizing the design strongly**.

Research Question 2: Which screens/labels (e.g., buttons) cause hesitation or errors that lower satisfaction?

Claim: The main sources of confusion and hesitation were

1. The **food-bank verification and food-bank vs. Grocery distinction**
2. The add-to-cart and quantity control
3. The category labels

Evidence: Based on [plot 3](#), most often mentioned as confusing are these areas

1. Verification /food-bank survey – 5/10 participants
Participants were unsure whether they were donating vs. receiving, what the survey was for, and who qualifies as “food insecure.”
2. Item quantity/add-to-cart - 5/10 participants
 - Participants complained that there is not strong feedback after tapping “add to cart.”
 - Confusion in quantity handling, difficulty remembering already selected items
 - This aligns with the closed-ended question where only 30% answered “yes” to “Was it clear to you when your selections were successfully added to the checkout bin?”
3. Categories—3/10 participants
They noted that categories were too wordy for requested icons or clearer placeholders.

Confidence:

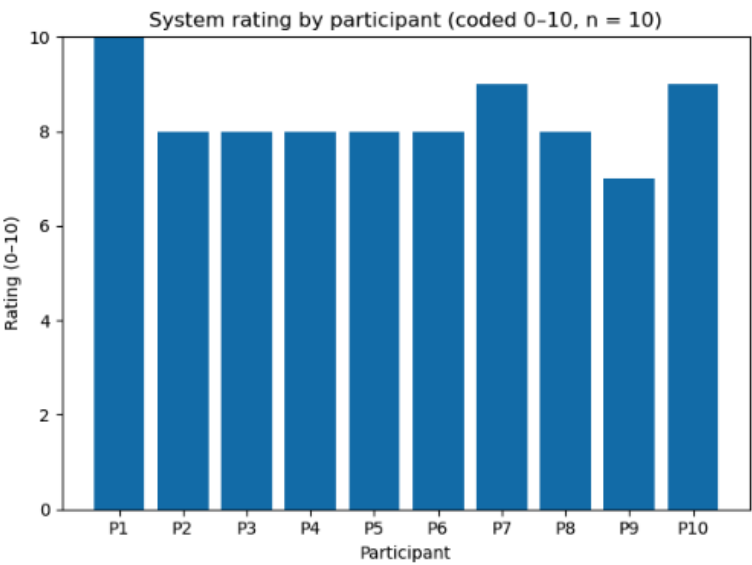
We are moderately confident about "3 main sources cause the confusion" because

- Strength:
 - The same **friction areas** appear in multiple participants' comments.
 - We use an observational study to capture the user's act, and a semi-structure interview to capture the reasoning and thoughts
- Limitation:
 - We used simple keyword-based coding to assign components into feature buckets. Since we are also the designers of the app, doing more **interpretive thematic analysis** would have a **high risk of researchers' bias**. We missed an independent researcher conducting a more detailed thematic analysis that might reveal subthemes within each friction area.
 - We did not collect page-level timing data; our conclusions about where users “hesitate” are based on self-reported difficulties and our observations, rather than the **objective timing measures**.

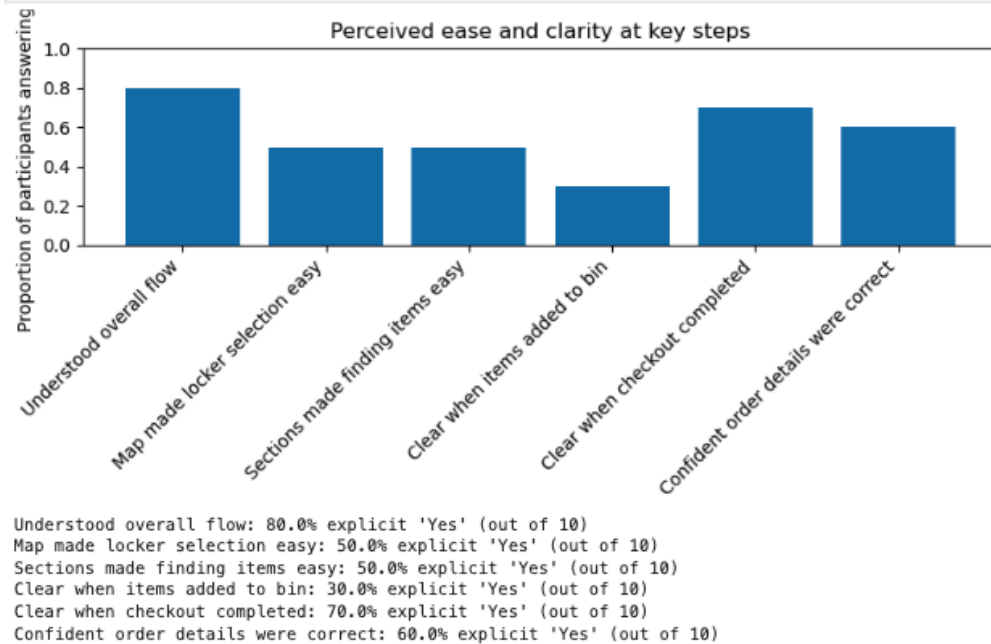
Visualizations (1 page max)

Final coded ratings (0-10) for all participants:

	participant_id	rating_10_final
5	P1	10
24	P2	8
42	P3	8
60	P4	8
79	P5	8
98	P6	8
117	P7	9
136	P8	8
155	P9	7
174	P10	9

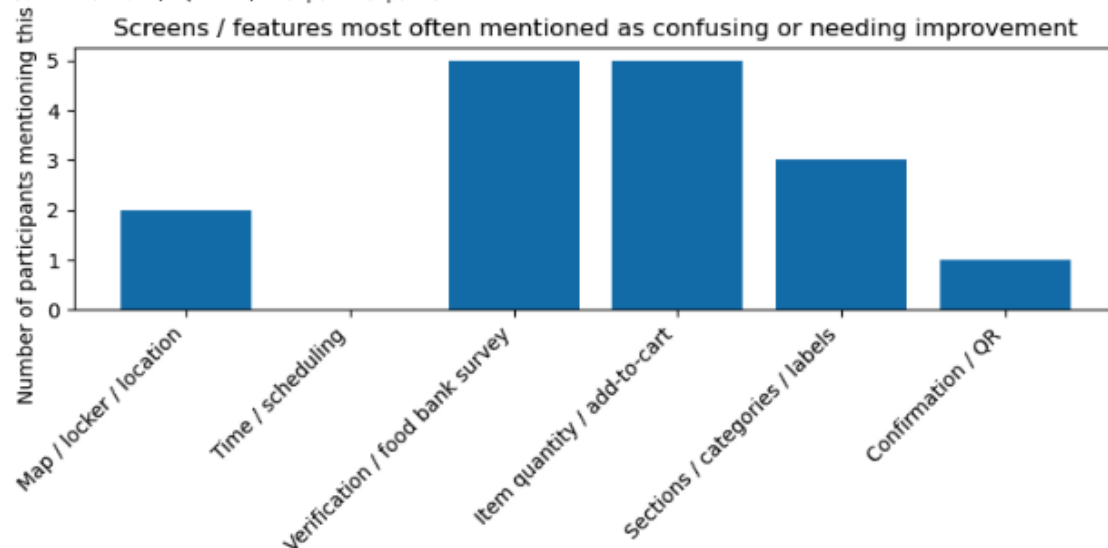


Plot 1 bar chart of the rating



Plot 2 bar chart of the binary voting for the subsection

Friction feature counts (participants mentioning each area):
Map / locker / location: 2 / 10 participants
Time / scheduling: 0 / 10 participants
Verification / food bank survey: 5 / 10 participants
Item quantity / add-to-cart: 5 / 10 participants
Sections / categories / labels: 3 / 10 participants
Confirmation / QR: 1 / 10 participants



Plot 3 Bar plot of count of subsection causing hesitation and confusion

Recommendations and Critique (1 page max)

i) Design recommendations:

1. Reflect on your interface design.

Our evaluation, supported by piloting and prototyping, shows that the core interaction flow: browse -> add items -> choose locker/time -> confirm, works effectively for first-time users. Nearly all participants completed the sequence without assistance and rated the experience positively ([plot 1](#)), indicating that the interface would likely function well in natural campus settings. Consistent styling, simplified navigation, and a clean aesthetic contributed to this smooth performance. Several areas nevertheless require refinement. Confusion between the grocery and food-bank streams persisted, with some participants still unsure whether they were donating or receiving food despite wording revisions. In addition, **70%** of users were uncertain when items were successfully added to the cart, reflecting insufficient micro-interaction feedback ([plot 2](#)). Category labels were also overly text-heavy, and participants expressed a preference for clearer iconography.

2. Recommend the next design step.

The next design step is to clarify the distinction between the grocery and food-bank flows by improving onboarding and redesigning the verification screen, since many participants were unsure whether they were donating or receiving food. This may include revisiting the labels and explanatory text used to introduce the two pathways, especially since our piloting revealed that earlier wording such as “Food Bank Pick Up” created misunderstandings even after revisions. Strengthening add-to-cart feedback, such as adding quantity selectors or visible confirmation cues, is also essential, given users’ uncertainty about whether items were successfully added. Simplifying category labels with clearer icons would further reduce the cognitive effort required to locate items. At the system level, future iterations should incorporate campus foot-traffic data to guide locker placement, ensuring accessibility while remaining cost-effective.

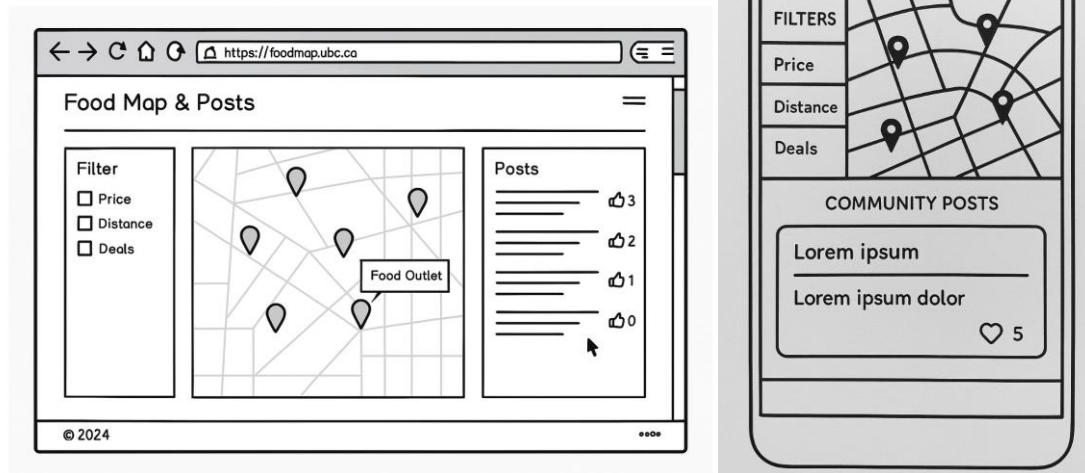
j) Critique of process:

Our evaluation methods effectively surfaced the main usability issues in our interface. The think-aloud protocol revealed real-time hesitation, especially around add-to-cart feedback and the food-bank verification survey, while semi-structured interviews helped explain why certain labels and screens were confusing. Piloting also proved valuable by identifying unclear wording and missing elements such as the payment flow. However, several factors limited how closely our process followed user-centred design principles. Explaining the app’s purpose before testing likely inflated task-completion success, and moderated sessions may have encouraged socially

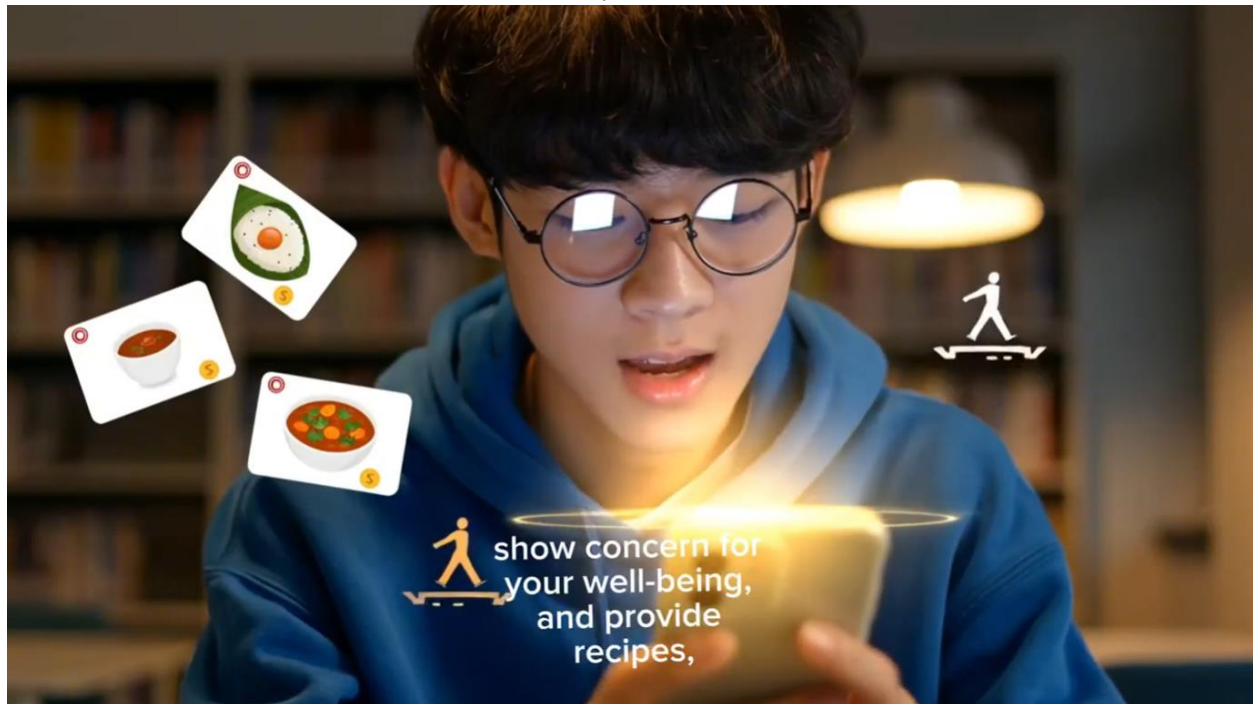
desirable responses. Our reliance on keyword-based coding introduced potential designer bias, and without timing or click-level metrics, hesitation was assessed only through observation. The most impactful improvement would be conducting contextual inquiry and testing low-fidelity prototypes earlier. This would have grounded our design in actual student behaviour, revealed conceptual misunderstandings, such as confusion between grocery and food-bank flows, much earlier, and produced more authentic feedback. Future work should also use lo-fi sketches, unmoderated sessions, and objective performance metrics.

Appendix A (Alternative Conceptual Model)

Google Map + Community Post



Chat Bot Food Advisor (Our conceptual model was visualized with video which can't be added to our report so added a scene from the video)



Appendix B (Prototype)

Link to prototype (Figma, Google Slides, etc etc etc.)

Figma: <https://www.figma.com/design/iEuqR15GXceiWvJpPXt5ky/Food-Bank-App-Prototype?node-id=32-482&t=qNqKaoWZ6d0Gfawg-1>

Google Slides:

https://docs.google.com/presentation/d/1rWDdUwA_PKZVdX_pdjxa0MiN9Kaq4nVb/edit?usp=s_haring&oid=101993433149874662345&rtpof=true&sd=true

Appendix C (Study Materials)

Study Materials

Raw or minimally aggregated data. Any content containing sensitive or participant data must remain on OneDrive (UBC server). To avoid access issues, please zip **participant generated data** and **submit separately**.

Content without sensitive or participant information can be stored on the platform of your choice.

[Data Collection](#)

[Data Analysis by UBC Jupyter notebook](#)

Contributions

Please summarize the individual contributions of each member and the total number of hours worked on this assignment.

Name	Contribution	Hours
Joseph Wray-Perez	Evaluation and Prototyping Methods c) and d)	7
Signature	Joseph Wray-Perez Date	05/12/2025
Eun Ji Hwang	Analysis and Data visualization	7
Signature	Eun Ji Hwang Date	05/12/2025
Wanying Zheng	Recommendations and Critique + Appendix	7
Signature	Wanying Zheng Date	05/12/2025
Jia Jung	Introduction + Conceptual Model Iterations	7
Signature	Jia Jung Date	05/12/2025
Hannah Wong	Evaluation and Prototyping Methods e) & f)	7
Signature	Hannah Wong Date	05/12/2025
Signature	Date	

AI Use Disclosure Form (Mandatory)

This disclosure form helps you reflect on and document your AI use to ensure you're leveraging these tools ethically to enhance your learning while developing the critical skills this course aims to teach. It's required for each deliverable (e.g., assignment, project report). If you did not use AI, please answer Q1.

1. Which AI tools did you use? (Check all that apply)

- Chatbot(s) (e.g., ChatGPT, Claude, Google Gemini)
- Coding Assistant(s) (e.g., Cursor.io, Copilot)
- Image Generator(s) (e.g., Midjourney, DALL-E, Stable Diffusion)
- Other(s): _____
- None

2. Purpose of Each AI Use (For each selected tool)

Briefly describe how the selected tools were used.*Estimate the proportion of the final submission that was directly shaped by AI output (0–100 %).

Tool(s) - Models (E.g., ChatGPT - GPT-5)	Task supported (list all that apply. E.g., ideation / writing / coding / analysis / visuals / other)	Description of Use (~1-2 sentences per task)	% of the work directly influenced*
Grammarly	Analysis	- Use to check the grammar error and keep the writing clear and simple	5
Gemini	Conceptual Model User Flow Picture	- Use to generate picture that shows user flow of vending machine conceptual model	2

3. Reflection on Responsible Use

How did you use AI as a learning tool without letting it do the work for you? For instance, describe:

- How AI helped you learn (not just complete the assignment),
- What steps you took to ensure you still developed the intended skills and did your own thinking,
- How you maintained academic integrity, and
- etc.

Include examples of prompts you used (these don't count toward the word count).

Suggested length: ~100 words per team for group projects; per student for individual work

Your Response

We mainly used AI throughout our project to help us articulate our points of view in a clear, concise, and grammatically correct way. We found throughout the entirety of this term-long project that AI is a very good tool at taking your ideas and articulating them very clearly. We learned how to make our thoughts clearer by looking at how AI would simplify our sentences and paragraphs. We ensured that we did our own thinking and developed our skills along the way by constantly checking in with course staff and discussing amongst the group. We held each other accountable for our parts and ensured that each person understood each facet of the assignment, in addition, we used AI to generate some images to help with our conceptual model. AI lets us personalize what we wanted to display as our conceptual model.